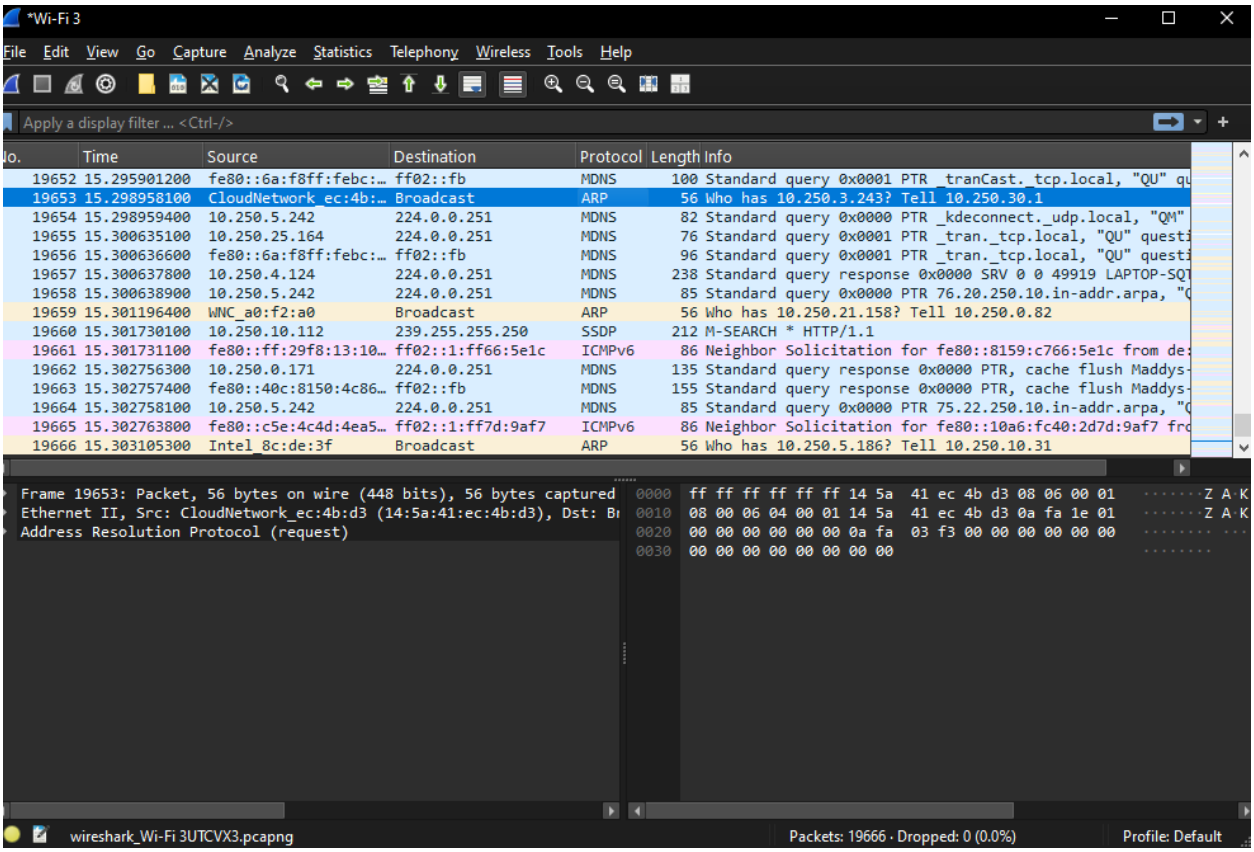
**CNNP – SKILL EXPERIMENTS**

**Skill 1 – Encapsulation Builder**

Aim: To capture network packets and understand how Ethernet, IP, and TCP/UDP headers are added during data transmission.

Step 1: Start Packet Capture in Wireshark




Step 2. Generate network traffic

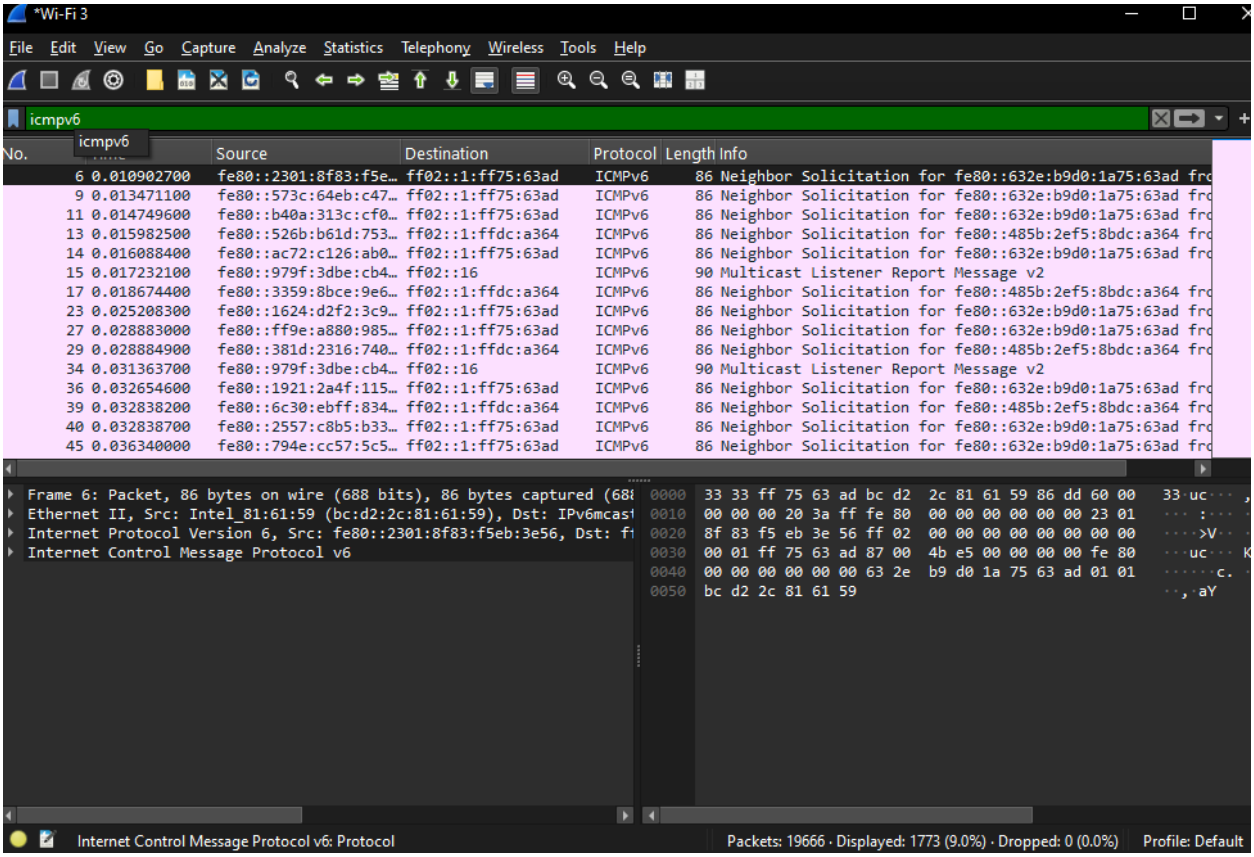
```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\GANESH>ping -n 4 8.8.8.8

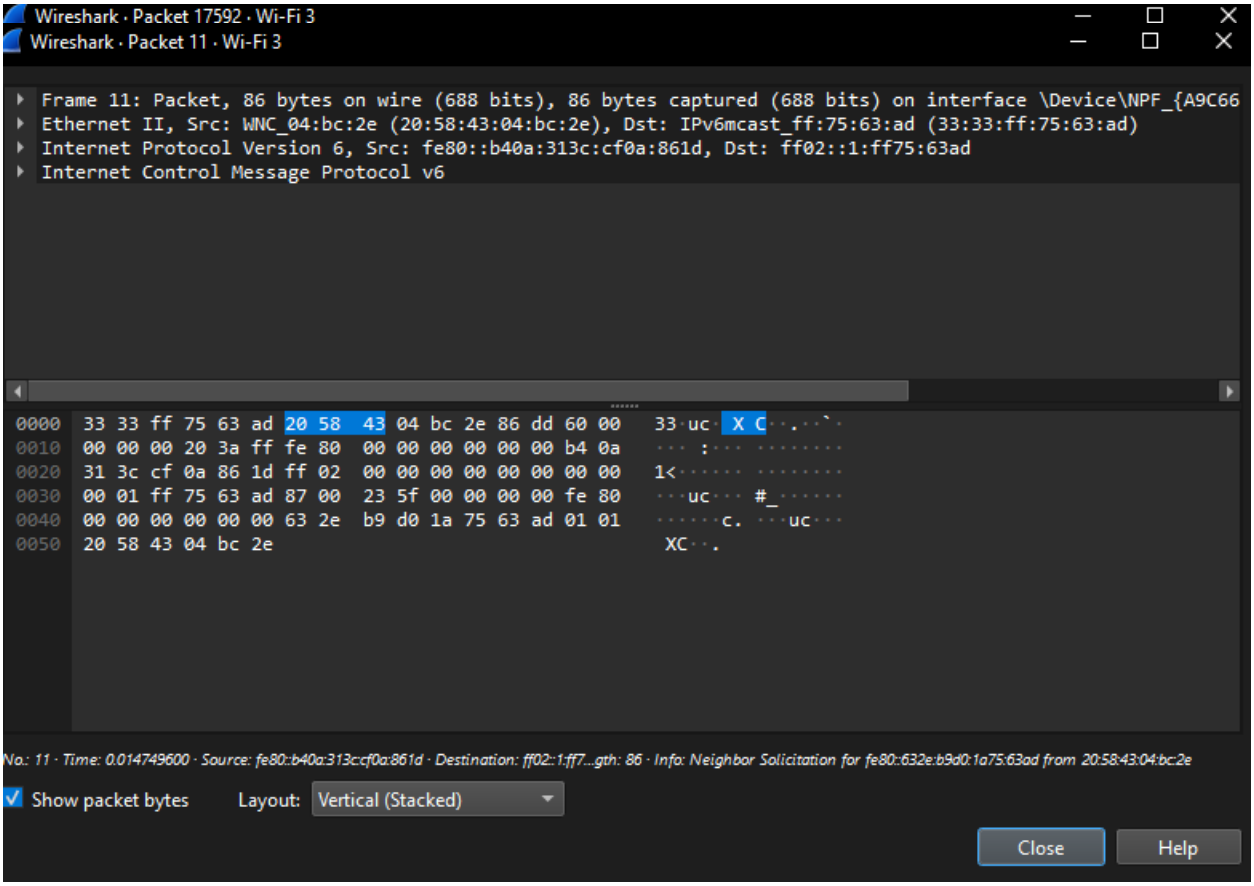
Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=43ms TTL=115
Reply from 8.8.8.8: bytes=32 time=27ms TTL=115
Reply from 8.8.8.8: bytes=32 time=75ms TTL=115
Reply from 8.8.8.8: bytes=32 time=24ms TTL=115

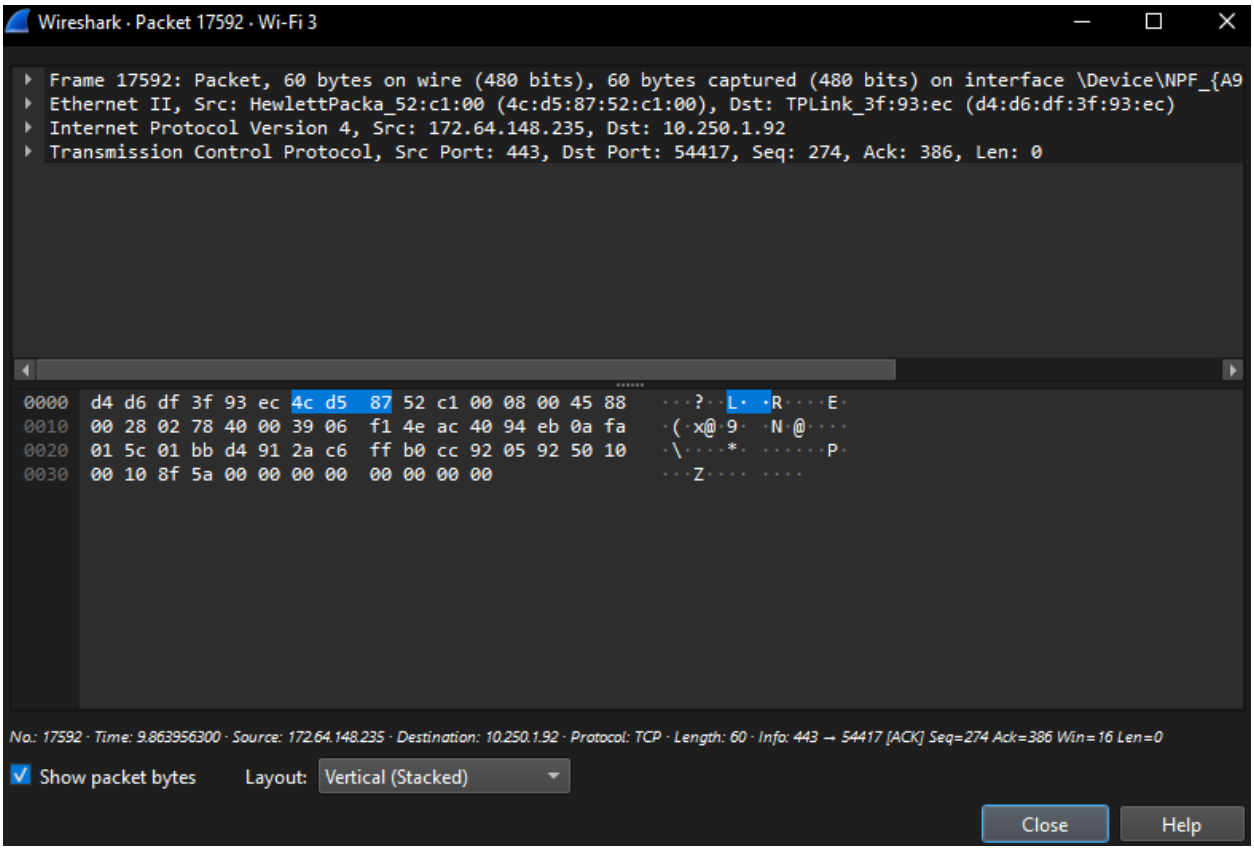
Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 24ms, Maximum = 75ms, Average = 42ms
```

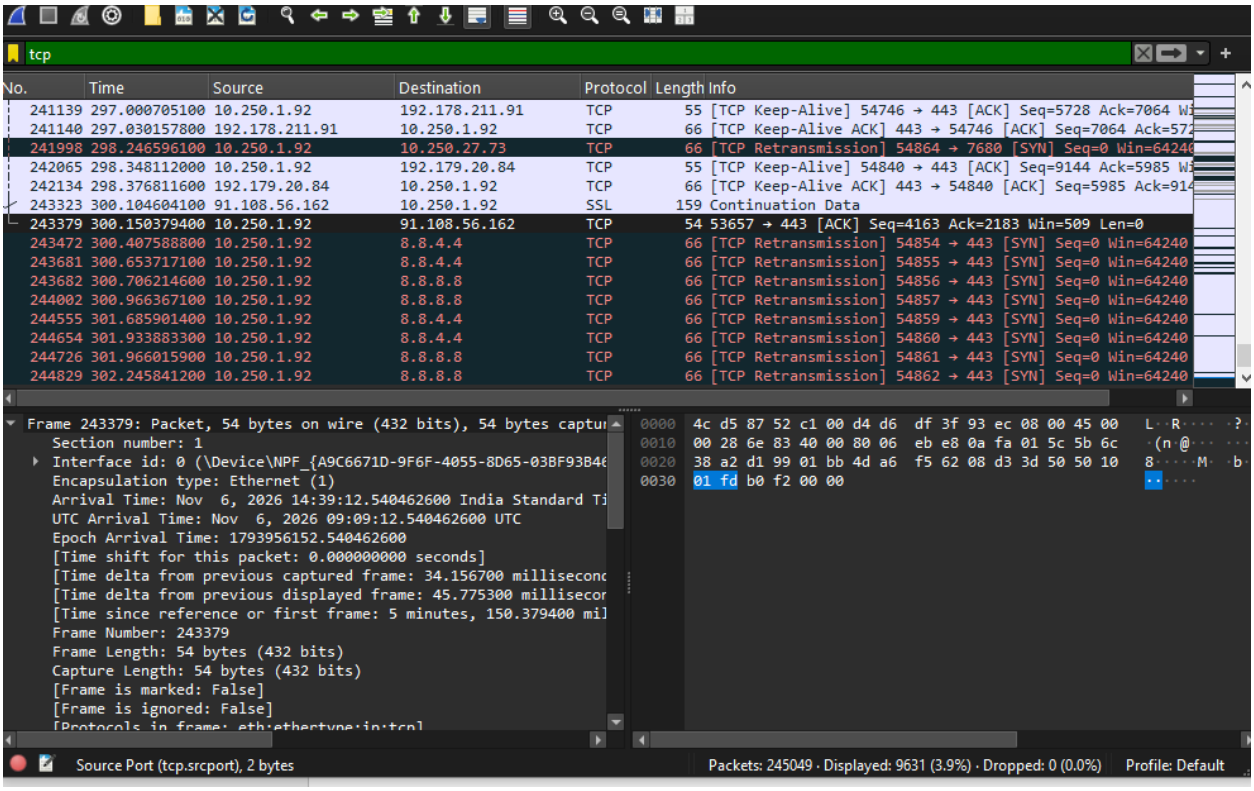
Step 3. Filter the captured packets

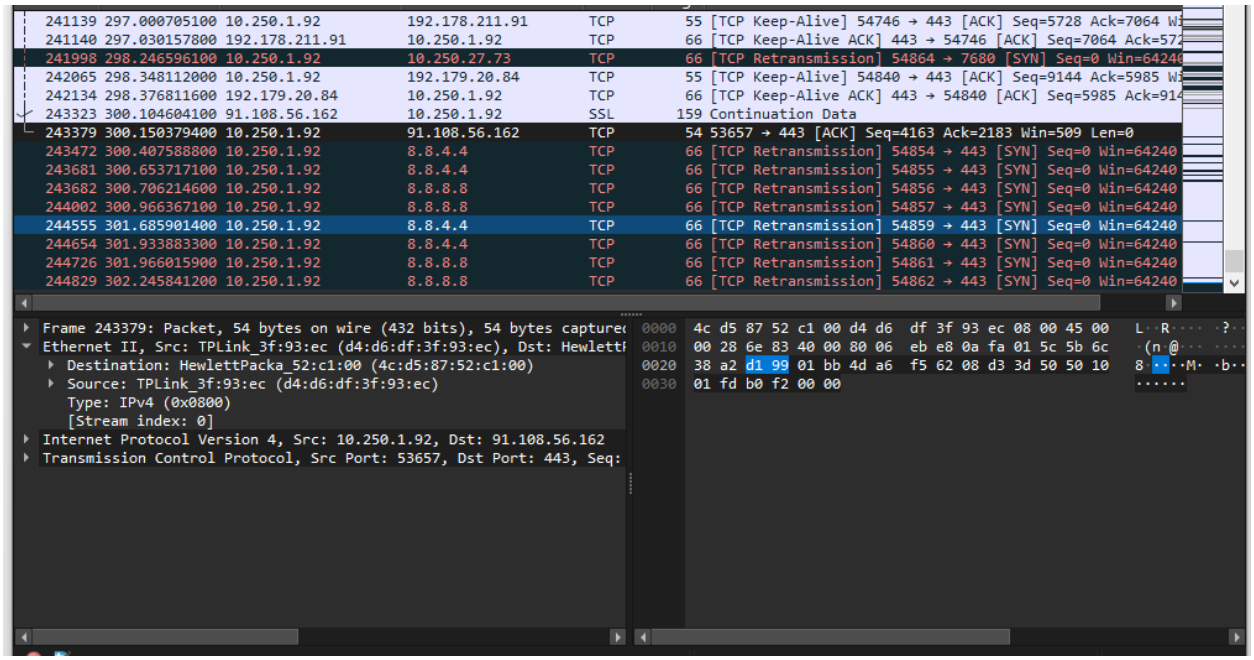


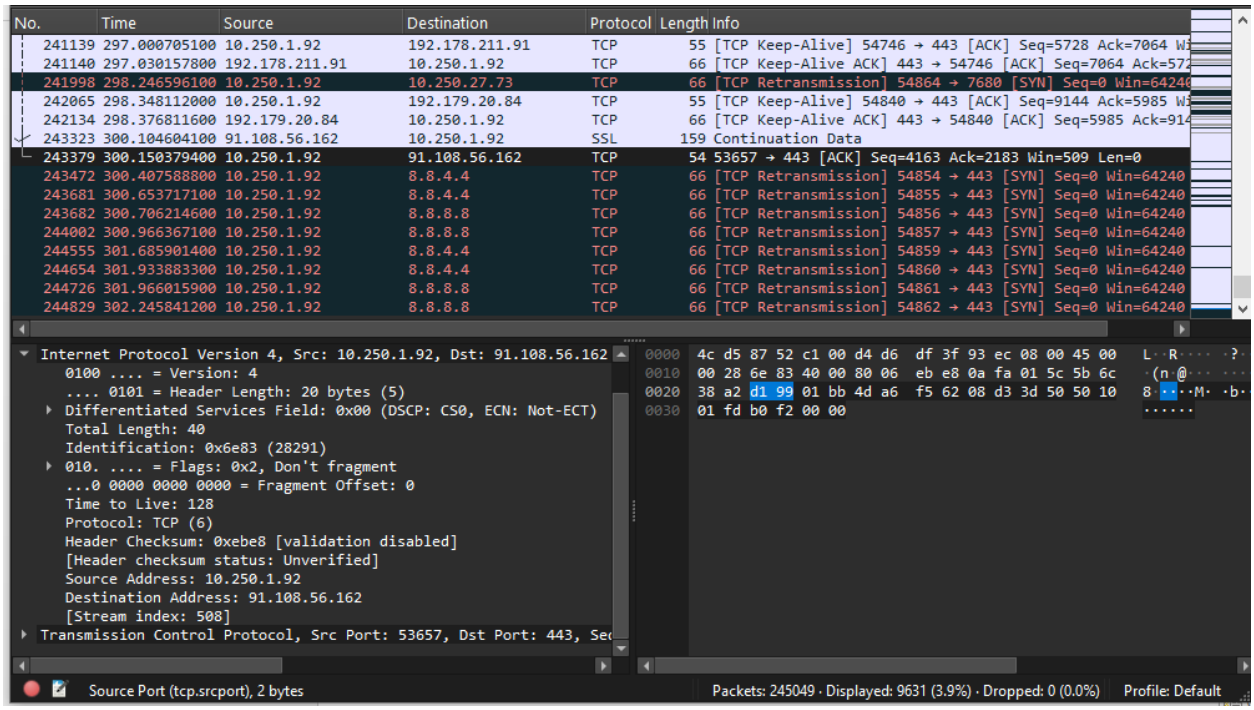
Step 4. Inspect encapsulation

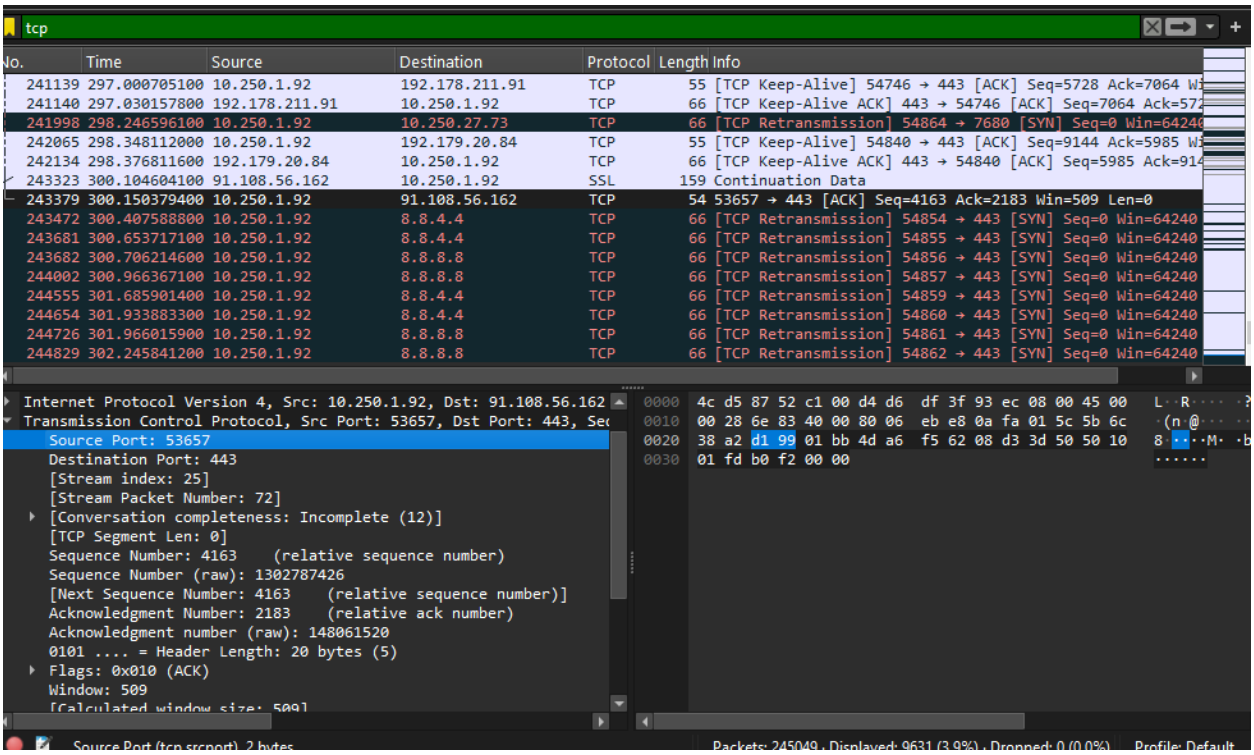



Wireshark · Packet 17592 · Wi-Fi 3
Frame 17592: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface \Device\NPF_{A9
Ethernet II, Src: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00), Dst: TPLink_3f:93:ec (d4:d6:df:3f:93:ec)
Internet Protocol Version 4, Src: 172.64.148.235, Dst: 10.250.1.92
Transmission Control Protocol, Src Port: 443, Dst Port: 54417, Seq: 274, Ack: 386, Len: 0
0000 d4 d6 df 3f 93 ec 4c d5 87 52 c1 00 08 00 45 88
0010 00 28 02 78 40 00 39 06 f1 4e ac 40 94 eb 0a fa
0020 01 5c 01 bb d4 91 2a c6 ff b0 cc 92 05 92 50 10
0030 00 10 8f 5a 00 00 00 00 00 00 00 00
No.: 17592 · Time: 9.863956300 · Source: 172.64.148.235 · Destination: 10.250.1.92 · Protocol: TCP · Length: 60 · Info: 443 → 54417 [ACK] Seq=274 Ack=386 Win=16 Len=0
Show packet bytes
Layout: Vertical (Stacked)
Close
Help


tcp
No. Time Source Destination Protocol Length Info
241139 297.000705100 10.250.1.92 192.178.211.91 TCP 55 [TCP Keep-Alive] 54746 → 443 [ACK] Seq=5728 Ack=7064
241140 297.030157800 192.178.211.91 10.250.1.92 TCP 66 [TCP Keep-Alive ACK] 443 → 54746 [ACK] Seq=7064 Ack=57
241998 298.246596100 10.250.1.92 10.250.27.73 TCP 66 [TCP Retransmission] 54864 → 7680 [SYN] Seq=0 Win=6424
242065 298.348112000 10.250.1.92 192.179.20.84 TCP 55 [TCP Keep-Alive] 54840 → 443 [ACK] Seq=9144 Ack=5985
242134 298.376811600 192.179.20.84 10.250.1.92 TCP 66 [TCP Keep-Alive ACK] 443 → 54840 [ACK] Seq=5985 Ack=914
243323 300.104604100 91.108.56.162 10.250.1.92 SSL 159 Continuation Data
243379 300.150379400 10.250.1.92 91.108.56.162 TCP 54 53657 → 443 [ACK] Seq=4163 Ack=2183 Win=509 Len=0
243472 300.407588800 10.250.1.92 8.8.4.4 TCP 66 [TCP Retransmission] 54854 → 443 [SYN] Seq=0 Win=64240
243681 300.653717100 10.250.1.92 8.8.4.4 TCP 66 [TCP Retransmission] 54855 → 443 [SYN] Seq=0 Win=64240
243682 300.706214600 10.250.1.92 8.8.8.8 TCP 66 [TCP Retransmission] 54856 → 443 [SYN] Seq=0 Win=64240
244002 300.966367100 10.250.1.92 8.8.8.8 TCP 66 [TCP Retransmission] 54857 → 443 [SYN] Seq=0 Win=64240
244555 301.685901400 10.250.1.92 8.8.4.4 TCP 66 [TCP Retransmission] 54859 → 443 [SYN] Seq=0 Win=64240
244654 301.933883300 10.250.1.92 8.8.4.4 TCP 66 [TCP Retransmission] 54860 → 443 [SYN] Seq=0 Win=64240
244726 301.966015900 10.250.1.92 8.8.8.8 TCP 66 [TCP Retransmission] 54861 → 443 [SYN] Seq=0 Win=64240
244829 302.245841200 10.250.1.92 8.8.8.8 TCP 66 [TCP Retransmission] 54862 → 443 [SYN] Seq=0 Win=64240
Frame 243379: Packet, 54 bytes on wire (432 bits), 54 bytes captur
Section number: 1
Interface id: 0 (\Device\NPF_{A9C6671D-9F6F-4055-8D65-03BF93B46
Encapsulation type: Ethernet (1)
Arrival Time: Nov 6, 2026 14:39:12.540462600 India Standard Ti
UTC Arrival Time: Nov 6, 2026 09:09:12.540462600 UTC
Epoch Arrival Time: 1793956152.540462600
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 34.156700 millisecond
[Time delta from previous displayed frame: 45.775300 millisecon
[Time since reference or first frame: 5 minutes, 150.379400 mil
Frame Number: 243379
Frame Length: 54 bytes (432 bits)
Capture Length: 54 bytes (432 bits)
[Frame is marked: False]
[Frame is ignored: False]
0000 4c d5 87 52 c1 00 d4 d6 df 3f 93 ec 08 00 45 00
0010 00 28 6e 83 40 00 80 06 eb e8 0a fa 01 5c 5b 6c
0020 38 a2 d1 99 01 bb 4d a6 f5 62 08 d3 3d 50 50 10
0030 01 fd b0 f2 00 00
Source Port (tcp.srcport), 2 bytes
Packets: 245049 · Displayed: 9631 (3.9%) · Dropped: 0 (0.0%)
Profile: Default


241139 297.000705100 10.250.1.92 192.178.211.91 TCP 55 [TCP Keep-Alive] 54746 → 443 [ACK] Seq=5728 Ack=7064
241140 297.030157800 192.178.211.91 10.250.1.92 TCP 66 [TCP Keep-Alive ACK] 443 → 54746 [ACK] Seq=7064 Ack=572
241998 298.246596100 10.250.1.92 10.250.27.73 TCP 66 [TCP Retransmission] 54864 → 7680 [SYN] Seq=0 Win=64240
242065 298.348112000 10.250.1.92 192.179.20.84 TCP 55 [TCP Keep-Alive] 54840 → 443 [ACK] Seq=9144 Ack=5985
242134 298.376811600 192.179.20.84 10.250.1.92 TCP 66 [TCP Keep-Alive ACK] 443 → 54840 [ACK] Seq=5985 Ack=914
243323 300.104604100 91.108.56.162 10.250.1.92 SSL 159 Continuation Data
243379 300.150379400 10.250.1.92 91.108.56.162 TCP 54 53657 → 443 [ACK] Seq=4163 Ack=2183 Win=509 Len=0
243472 300.407588800 10.250.1.92 8.8.4.4 TCP 66 [TCP Retransmission] 54854 → 443 [SYN] Seq=0 Win=64240
243681 300.653717100 10.250.1.92 8.8.4.4 TCP 66 [TCP Retransmission] 54855 → 443 [SYN] Seq=0 Win=64240
243682 300.706214600 10.250.1.92 8.8.8.8 TCP 66 [TCP Retransmission] 54856 → 443 [SYN] Seq=0 Win=64240
244002 300.966367100 10.250.1.92 8.8.8.8 TCP 66 [TCP Retransmission] 54857 → 443 [SYN] Seq=0 Win=64240
244555 301.685901400 10.250.1.92 8.8.4.4 TCP 66 [TCP Retransmission] 54859 → 443 [SYN] Seq=0 Win=64240
244654 301.933883300 10.250.1.92 8.8.4.4 TCP 66 [TCP Retransmission] 54860 → 443 [SYN] Seq=0 Win=64240
244726 301.966015900 10.250.1.92 8.8.8.8 TCP 66 [TCP Retransmission] 54861 → 443 [SYN] Seq=0 Win=64240
244829 302.245841200 10.250.1.92 8.8.8.8 TCP 66 [TCP Retransmission] 54862 → 443 [SYN] Seq=0 Win=64240
Frame 243379: Packet, 54 bytes on wire (432 bits), 54 bytes capture
Ethernet II, Src: TPLink_3f:93:ec (d4:d6:df:3f:93:ec), Dst: Hewlett
Destination: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
Source: TPLink_3f:93:ec (d4:d6:df:3f:93:ec)
Type: IPv4 (0x0800)
[Stream index: 0]
Internet Protocol Version 4, Src: 10.250.1.92, Dst: 91.108.56.162
Transmission Control Protocol, Src Port: 53657, Dst Port: 443, Seq:
0000 4c d5 87 52 c1 00 d4 d6 df 3f 93 ec 08 00 45 00
0010 00 28 6e 83 40 00 80 06 eb e8 0a fa 01 5c 5b 6c
0020 38 a2 d1 99 01 bb 4d a6 f5 62 08 d3 3d 50 50 10
0030 01 fd b0 f2 00 00


No. Time Source Destination Protocol Length Info
241139 297.000705100 10.250.1.92 192.178.211.91 TCP 55 [TCP Keep-Alive] 54746 → 443 [ACK] Seq=5728 Ack=7064
241140 297.030157800 192.178.211.91 10.250.1.92 TCP 66 [TCP Keep-Alive ACK] 443 → 54746 [ACK] Seq=7064 Ack=572
241998 298.246596100 10.250.1.92 10.250.27.73 TCP 66 [TCP Retransmission] 54864 → 7680 [SYN] Seq=0 Win=64240
242065 298.348112000 10.250.1.92 192.179.20.84 TCP 55 [TCP Keep-Alive] 54840 → 443 [ACK] Seq=9144 Ack=5985
242134 298.376811600 192.179.20.84 10.250.1.92 TCP 66 [TCP Keep-Alive ACK] 443 → 54840 [ACK] Seq=5985 Ack=914
243323 300.104604100 91.108.56.162 10.250.1.92 SSL 159 Continuation Data
243379 300.150379400 10.250.1.92 91.108.56.162 TCP 54 53657 → 443 [ACK] Seq=4163 Ack=2183 Win=509 Len=0
243472 300.407588800 10.250.1.92 8.8.4.4 TCP 66 [TCP Retransmission] 54854 → 443 [SYN] Seq=0 Win=64240
243681 300.653717100 10.250.1.92 8.8.4.4 TCP 66 [TCP Retransmission] 54855 → 443 [SYN] Seq=0 Win=64240
243682 300.706214600 10.250.1.92 8.8.8.8 TCP 66 [TCP Retransmission] 54856 → 443 [SYN] Seq=0 Win=64240
244002 300.966367100 10.250.1.92 8.8.8.8 TCP 66 [TCP Retransmission] 54857 → 443 [SYN] Seq=0 Win=64240
244555 301.685901400 10.250.1.92 8.8.4.4 TCP 66 [TCP Retransmission] 54859 → 443 [SYN] Seq=0 Win=64240
244654 301.933883300 10.250.1.92 8.8.4.4 TCP 66 [TCP Retransmission] 54860 → 443 [SYN] Seq=0 Win=64240
244726 301.966015900 10.250.1.92 8.8.8.8 TCP 66 [TCP Retransmission] 54861 → 443 [SYN] Seq=0 Win=64240
244829 302.245841200 10.250.1.92 8.8.8.8 TCP 66 [TCP Retransmission] 54862 → 443 [SYN] Seq=0 Win=64240
Internet Protocol Version 4, Src: 10.250.1.92, Dst: 91.108.56.162
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 40
Identification: 0x6e83 (28291)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 128
Protocol: TCP (6)
Header Checksum: 0xebe8 [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.250.1.92
Destination Address: 91.108.56.162
[Stream index: 508]
Transmission Control Protocol, Src Port: 53657, Dst Port: 443, Se
0000 4c d5 87 52 c1 00 d4 d6 df 3f 93 ec 08 00 45 00
0010 00 28 6e 83 40 00 80 06 eb e8 0a fa 01 5c 5b 6c
0020 38 a2 d1 99 01 bb 4d a6 f5 62 08 d3 3d 50 50 10
0030 01 fd b0 f2 00 00
Source Port (tcp.srcport), 2 bytes
Packets: 245049 · Displayed: 9631 (3.9%) · Dropped: 0 (0.0%)
Profile: Default

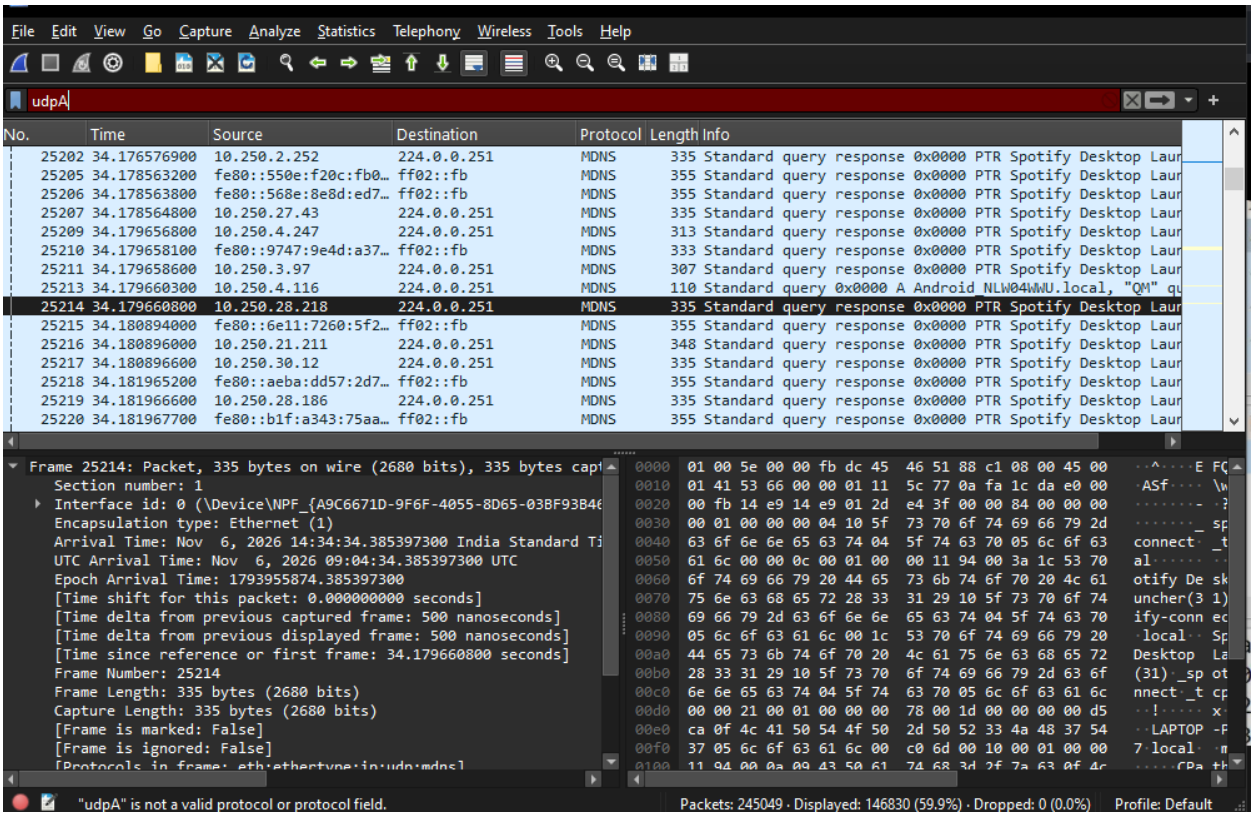


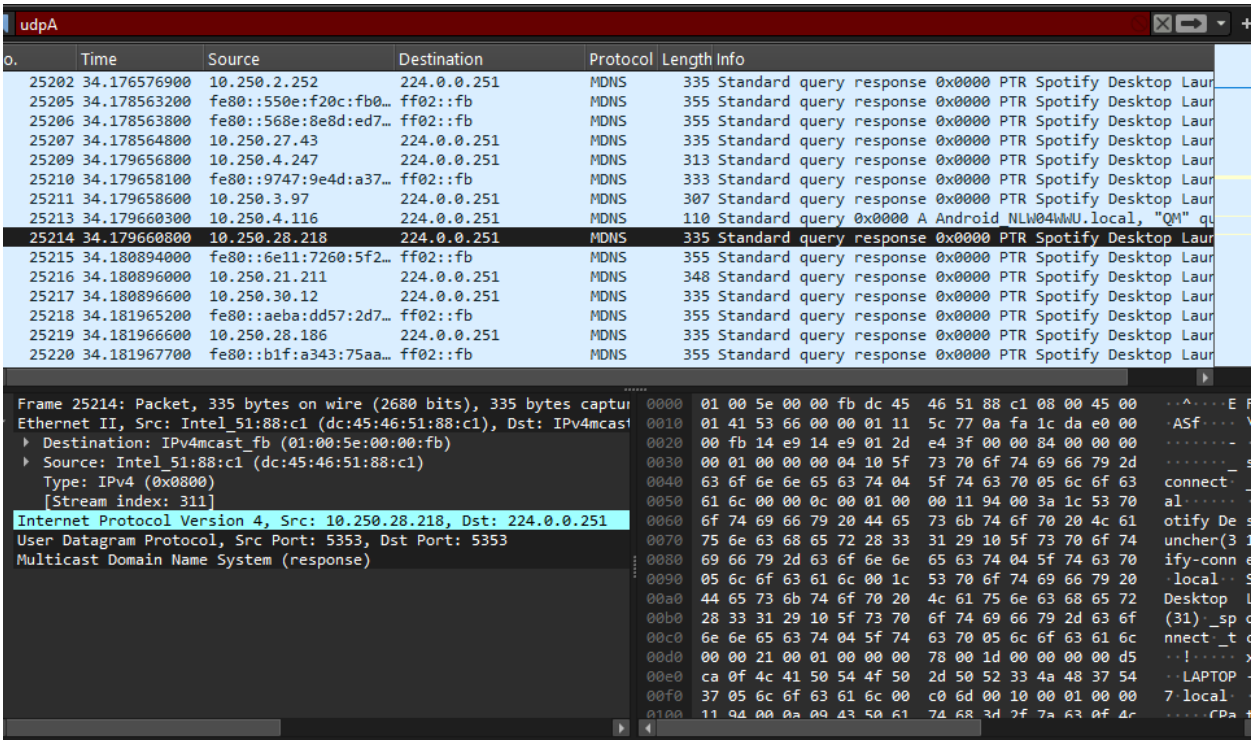

## TCP observation

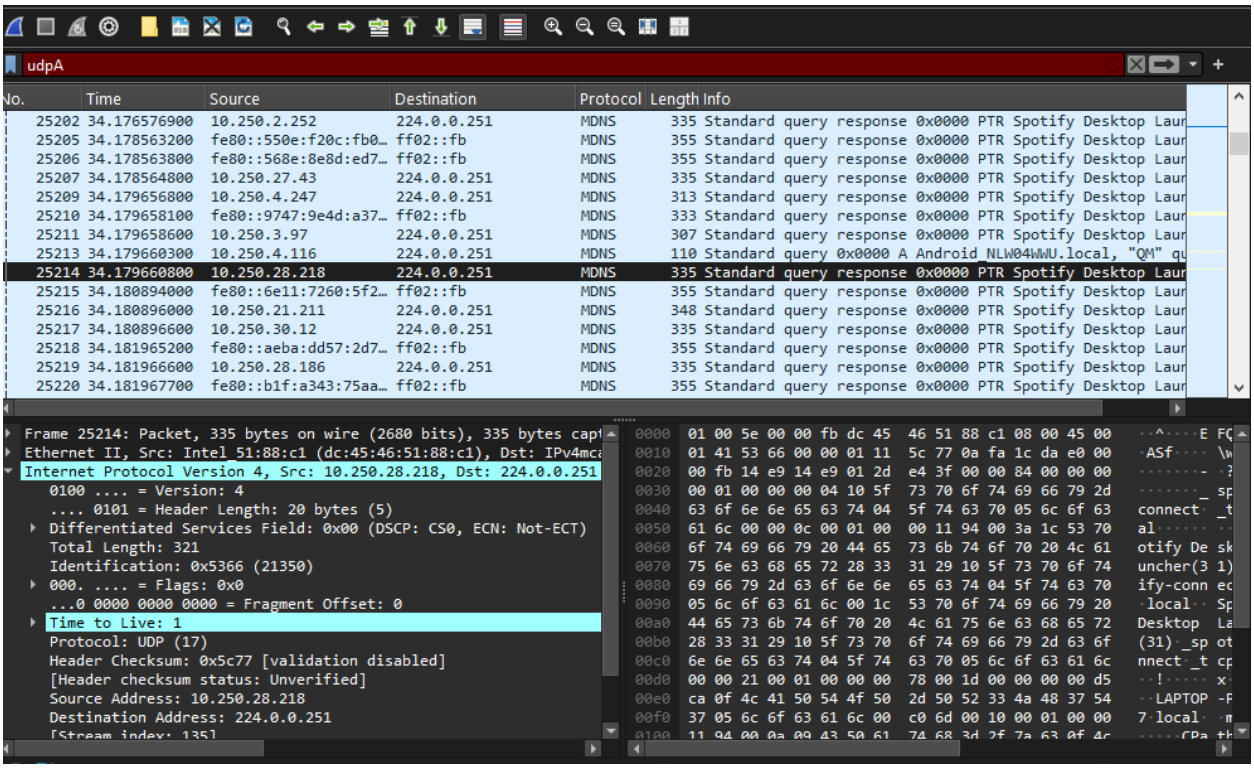
Selected packet is No. 243379.

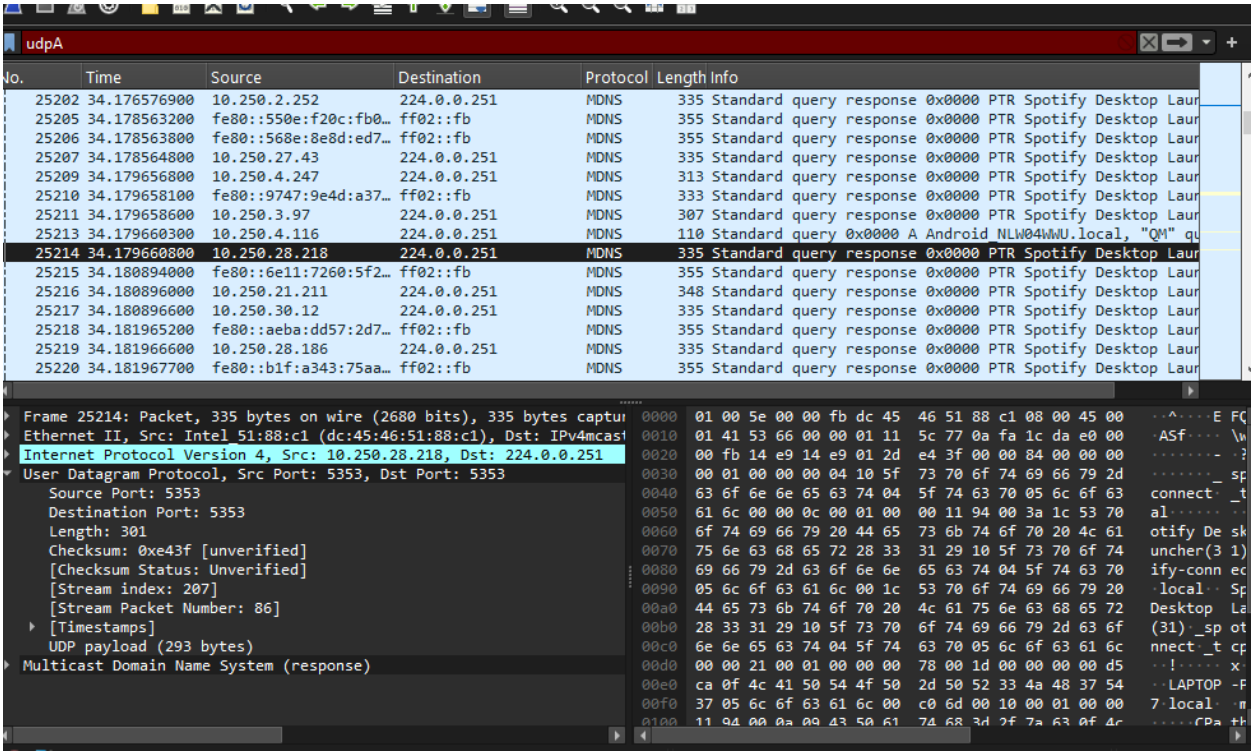
| Parameter | Observed value |
|---|---|
| Frame Length | 54 bytes |
| Ethernet Header Length | 14 bytes |
| Source MAC | d4:d6:df:3f:93:ec |
| Destination MAC | 4c:d5:87:52:c1:00 |
| IPv4 Header Length | 20 bytes |
| IPv4 Total Length | 40 bytes |
| TCP Header Length | 20 bytes |
| TCP Segment Length | 0 bytes |
| Application Payload | 0 bytes |

observation: This is a TCP ACK packet without application data. That's why its TCP segment length is 0.


File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udpA
No. Time Source Destination Protocol Length Info
25202 34.176576900 10.250.2.252 224.0.0.251 MDNS 335 Standard query response 0x0000 PTR Spotify Desktop Laur
25205 34.178563200 fe80::550e:f20c:fb0… ff02::fb MDNS 355 Standard query response 0x0000 PTR Spotify Desktop Laur
25206 34.178563800 fe80::568e:8e8d:ed7… ff02::fb MDNS 355 Standard query response 0x0000 PTR Spotify Desktop Laur
25207 34.178564800 10.250.27.43 224.0.0.251 MDNS 335 Standard query response 0x0000 PTR Spotify Desktop Laur
25209 34.179656800 10.250.4.247 224.0.0.251 MDNS 313 Standard query response 0x0000 PTR Spotify Desktop Laur
25210 34.179658100 fe80::9747:9e4d:a37… ff02::fb MDNS 333 Standard query response 0x0000 PTR Spotify Desktop Laur
25211 34.179658600 10.250.3.97 224.0.0.251 MDNS 307 Standard query response 0x0000 PTR Spotify Desktop Laur
25213 34.179660300 10.250.4.116 224.0.0.251 MDNS 110 Standard query 0x0000 A Android_NLW04WWU.local, "QM" qu
25214 34.179660800 10.250.28.218 224.0.0.251 MDNS 335 Standard query response 0x0000 PTR Spotify Desktop Laur
25215 34.180894000 fe80::6e11:7260:5f2… ff02::fb MDNS 355 Standard query response 0x0000 PTR Spotify Desktop Laur
25216 34.180896000 10.250.21.211 224.0.0.251 MDNS 348 Standard query response 0x0000 PTR Spotify Desktop Laur
25217 34.180896600 10.250.30.12 224.0.0.251 MDNS 335 Standard query response 0x0000 PTR Spotify Desktop Laur
25218 34.181965200 fe80::aeba:dd57:2d7… ff02::fb MDNS 355 Standard query response 0x0000 PTR Spotify Desktop Laur
25219 34.181966600 10.250.28.186 224.0.0.251 MDNS 335 Standard query response 0x0000 PTR Spotify Desktop Laur
25220 34.181967700 fe80::b1f:a343:75aa… ff02::fb MDNS 355 Standard query response 0x0000 PTR Spotify Desktop Laur
Frame 25214: Packet, 335 bytes on wire (2680 bits), 335 bytes capt
Section number: 1
Interface id: 0 (\Device\NPF_{A9C6671D-9F6F-4055-8D65-03BF93B46
Encapsulation type: Ethernet (1)
Arrival Time: Nov 6, 2026 14:34:34.385397300 India Standard Ti
UTC Arrival Time: Nov 6, 2026 09:04:34.385397300 UTC
Epoch Arrival Time: 1793955874.385397300
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 500 nanoseconds]
[Time delta from previous displayed frame: 500 nanoseconds]
[Time since reference or first frame: 34.179660800 seconds]
Frame Number: 25214
Frame Length: 335 bytes (2680 bits)
Capture Length: 335 bytes (2680 bits)
[Frame is marked: False]
[Frame is ignored: False]
0000 01 00 5e 00 00 fb dc 45 46 51 88 c1 08 00 45 00
0010 01 41 53 66 00 00 01 11 5c 77 0a fa 1c da e0 00
0020 00 fb 14 e9 14 e9 01 2d e4 3f 00 00 84 00 00 00
0030 00 01 00 00 00 04 10 5f 73 70 6f 74 69 66 79 2d
0040 63 6f 6e 6e 65 63 74 04 5f 74 63 70 05 6c 6f 63
0050 61 6c 00 00 0c 00 01 00 00 11 94 00 3a 1c 53 70
0060 6f 74 69 66 79 20 44 65 73 6b 74 6f 70 20 4c 61
0070 75 6e 63 68 65 72 28 33 31 29 10 5f 73 70 6f 74
0080 69 66 79 2d 63 6f 6e 6e 65 63 74 04 5f 74 63 70
0090 05 6c 6f 63 61 6c 00 1c 53 70 6f 74 69 66 79 20
00a0 44 65 73 6b 74 6f 70 20 4c 61 75 6e 63 68 65 72
00b0 28 33 31 29 10 5f 73 70 6f 74 69 66 79 2d 63 6f
00c0 6e 6e 65 63 74 04 5f 74 63 70 05 6c 6f 63 61 6c
00d0 00 00 21 00 01 00 00 00 78 00 1d 00 00 00 00 d5
00e0 ca 0f 4c 41 50 54 4f 50 2d 50 52 33 4a 48 37 54
00f0 37 05 6c 6f 63 61 6c 00 c0 6d 00 10 00 01 00 00
"udpA" is not a valid protocol or protocol field.
Packets: 245049 · Displayed: 146830 (59.9%) · Dropped: 0 (0.0%)
Profile: Default


udpA
o. Time Source Destination Protocol Length Info
25202 34.176576900 10.250.2.252 224.0.0.251 MDNS 335 Standard query response 0x0000 PTR Spotify Desktop Laur
25205 34.178563200 fe80::550e:f20c:fb0… ff02::fb MDNS 355 Standard query response 0x0000 PTR Spotify Desktop Laur
25206 34.178563800 fe80::568e:8e8d:ed7… ff02::fb MDNS 355 Standard query response 0x0000 PTR Spotify Desktop Laur
25207 34.178564800 10.250.27.43 224.0.0.251 MDNS 335 Standard query response 0x0000 PTR Spotify Desktop Laur
25209 34.179656800 10.250.4.247 224.0.0.251 MDNS 313 Standard query response 0x0000 PTR Spotify Desktop Laur
25210 34.179658100 fe80::9747:9e4d:a37… ff02::fb MDNS 333 Standard query response 0x0000 PTR Spotify Desktop Laur
25211 34.179658600 10.250.3.97 224.0.0.251 MDNS 307 Standard query response 0x0000 PTR Spotify Desktop Laur
25213 34.179660300 10.250.4.116 224.0.0.251 MDNS 110 Standard query 0x0000 A Android_NLW04WWU.local, "QM" qu
25214 34.179660800 10.250.28.218 224.0.0.251 MDNS 335 Standard query response 0x0000 PTR Spotify Desktop Laur
25215 34.180894000 fe80::6e11:7260:5f2… ff02::fb MDNS 355 Standard query response 0x0000 PTR Spotify Desktop Laur
25216 34.180896000 10.250.21.211 224.0.0.251 MDNS 348 Standard query response 0x0000 PTR Spotify Desktop Laur
25217 34.180896600 10.250.30.12 224.0.0.251 MDNS 335 Standard query response 0x0000 PTR Spotify Desktop Laur
25218 34.181965200 fe80::aeba:dd57:2d7… ff02::fb MDNS 355 Standard query response 0x0000 PTR Spotify Desktop Laur
25219 34.181966600 10.250.28.186 224.0.0.251 MDNS 335 Standard query response 0x0000 PTR Spotify Desktop Laur
25220 34.181967700 fe80::b1f:a343:75aa… ff02::fb MDNS 355 Standard query response 0x0000 PTR Spotify Desktop Laur
Frame 25214: Packet, 335 bytes on wire (2680 bits), 335 bytes captur
Ethernet II, Src: Intel_51:88:c1 (dc:45:46:51:88:c1), Dst: IPv4mcast
Destination: IPv4mcast_fb (01:00:5e:00:00:fb)
Source: Intel_51:88:c1 (dc:45:46:51:88:c1)
Type: IPv4 (0x0800)
[Stream index: 311]
Internet Protocol Version 4, Src: 10.250.28.218, Dst: 224.0.0.251
User Datagram Protocol, Src Port: 5353, Dst Port: 5353
Multicast Domain Name System (response)
0000 01 00 5e 00 00 fb dc 45 46 51 88 c1 08 00 45 00
0010 01 41 53 66 00 00 01 11 5c 77 0a fa 1c da e0 00
0020 00 fb 14 e9 14 e9 01 2d e4 3f 00 00 84 00 00 00
0030 00 01 00 00 00 04 10 5f 73 70 6f 74 69 66 79 2d
0040 63 6f 6e 6e 65 63 74 04 5f 74 63 70 05 6c 6f 63
0050 61 6c 00 00 0c 00 01 00 00 11 94 00 3a 1c 53 70
0060 6f 74 69 66 79 20 44 65 73 6b 74 6f 70 20 4c 61
0070 75 6e 63 68 65 72 28 33 31 29 10 5f 73 70 6f 74
0080 69 66 79 2d 63 6f 6e 6e 65 63 74 04 5f 74 63 70
0090 05 6c 6f 63 61 6c 00 1c 53 70 6f 74 69 66 79 20
00a0 44 65 73 6b 74 6f 70 20 4c 61 75 6e 63 68 65 72
00b0 28 33 31 29 10 5f 73 70 6f 74 69 66 79 2d 63 6f
00c0 6e 6e 65 63 74 04 5f 74 63 70 05 6c 6f 63 61 6c
00d0 00 00 21 00 01 00 00 00 78 00 1d 00 00 00 00 d5
00e0 ca 0f 4c 41 50 54 4f 50 2d 50 52 33 4a 48 37 54
00f0 37 05 6c 6f 63 61 6c 00 c0 6d 00 10 00 01 00 00




## UDP observation

Your selected packet is No. 25214 (mDNS response).

UDP values extracted

| Parameter | Observed value |
|---|---|
| Frame Length | 335 bytes |

| | |
|---|---|
| Ethernet Header Length | 14 bytes |
| IPv4 Header Length | 20 bytes |
| IPv4 Total Length | 321 bytes |
| UDP Header Length | 8 bytes |
| UDP Length | 301 bytes |
| UDP Payload Length | 293 bytes |
| Source Port | 5353 |
| Destination Port | 5353 |

Important observation:

The selected packet is an mDNS response using UDP. The UDP header is 8 bytes, and the UDP payload is 293 bytes.

## TCP vs UDP comparison

Add this table:

| Feature | TCP | UDP |
|---|---|---|
| Connection | Connection-oriented | Connectionless |
| Header size | Minimum 20 bytes | 8 bytes |
| Reliability | Reliable | No delivery guarantee |
| Acknowledgment | Supported | Not supported |
| Speed | Generally slower | Generally faster |
| Example | HTTPS | DNS, mDNS |

## Actual UDP overhead calculation

From your captured packet:

- Ethernet header = 14 bytes
- IPv4 header = 20 bytes
- UDP header = 8 bytes
- UDP payload = 293 bytes

Total frame size:

14+20+8+293=335 bytes

Total overhead:

14+20+8= 42 bytes

Payload efficiency:

87.46%

## Questions and Answers

**1. What is encapsulation?**

Encapsulation is the process of adding protocol headers to data as it moves through different layers of a network. Ethernet, IP and TCP/UDP headers are added before transmission.

**2. What is the purpose of the Ethernet header?**

The Ethernet header contains source and destination MAC addresses and the Ethernet type field. It helps deliver frames within a local network.

**3. What information is stored in the IP header?**

The IP header contains:

- Source IP address
- Destination IP address
- IP version
- Header length
- Total length
- Time To Live (TTL)
- Protocol information

**4. What is the difference between TCP and UDP headers?**

TCP has a minimum header size of 20 bytes and contains sequence numbers, acknowledgment numbers, flags and window information. UDP has a fixed header size of 8 bytes containing source port, destination port, length and checksum.

**5. Why does each protocol add a header?**

Each protocol adds a header to provide the necessary information for communication, addressing, routing, delivery and error detection.

**6. What is packet overhead?**

Packet overhead is the additional number of bytes used by protocol headers and other control information apart from the actual application data.

**7. How does header overhead affect network efficiency?**

Header overhead increases the total packet size and consumes additional network bandwidth. Higher overhead reduces the percentage of bandwidth available for actual application data.

## Result

The experiment was successfully performed using Wireshark. Network packets were captured and Ethernet, IP, TCP and UDP headers were examined. Header overhead and payload efficiency were calculated to understand how protocol encapsulation affects network efficiency.